

Technology Stack

Date	26 March 2026
Project Name	Smart Lender

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Chosen to create a responsive, user-friendly, and interactive web interface with faster page development and cross-browser compatibility.
2	Backend / Server-Side	Python (Flask Framework)	Chosen because Flask is lightweight, easy to develop, and integrates efficiently with machine learning models for fast prediction processing.
3	Database / Data Storage	MySQL	Chosen to securely store user information, loan application details, and prediction history with reliable relational data management.
4	Cloud / Hosting / Deployment	Render (<i>or Local Flask Server if not deployed</i>)	Chosen for simple deployment, reliable hosting, and easy access to the web application from any device.
5	Version Control & CI/CD	Git & GitHub	Chosen for version control, code management, collaboration, and project

			history throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, XGBoost	Used for data preprocessing, machine learning model training, prediction generation, and performance evaluation of the loan approval system.